

Mounting Directive for Vans



Change Information 01/2025
from 04/08/2025

Supplement to the Van Mounting Directives

- Mounting Directives for the Sprinter – BR 907, Edition AeJ2025-1a
- Mounting Directives for the Sprinter – BR 910, Edition AeJ2025-1a
- Supplemental Directive for the eSprinter– BR 907, Edition AeJ2025-1a

Mercedes-Benz



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1.1 Scope of application

This change information 01/2025 is a supplement to the Mercedes-Benz AG van mounting directives. Upon publication, it becomes an integral part of the following mounting directives and is therefore binding for body-builder manufacturers to apply and comply with.

Model series	Directive	Edition
907	Mounting Directive for the Sprinter Model Series 907 (Rear/All-Wheel Drive)	AeJ2025-1a
910	Mounting Directive for the Sprinter Model Series 910 (Front Wheel Drive)	AeJ2025-1a
907	Supplemental Directive for the eSprinter – Model Series 907 (Rear Wheel Drive) in addition to the Mounting Directive for the Sprinter Model Series 907	AeJ2025-1a

1.2 Changes

Edition	Date	Chapter	Change description
01/2025	04/08/2025	3.3	Selecting the basic vehicle: Vibration absorber for open model designation – only relevant for eSprinter model series 907
		8.4.8	Current collection: Electric power take-off (ePTO, code E2C) – only relevant for eSprinter model series 907
		8.9.9	Reversing camera: New installation specifications for retrofitting digital reversing camera with software update for pre-installation code FR7 – relevant for all open model designations of model series 907/910 Note: For a complete description of the reversing camera, the entire chapter is listed.

1.3 Concept of this change information

1.3.1 General information

The following content will be incorporated with the next update of the mounting directive for a van model series – see Chapter 1.1 Scope of application (→ page 3) for current editions. This change information is therefore no longer required after publication of the new version of the mounting directives for this model series.

The document may be used only as a complete document; excerpts are not permitted.

The facts and information described here apply exclusively in combination with the documents mentioned in Chapter 1.1 Scope of application (→ page 3) and supplement them.

Vehicle modifications by the bodybuilder manufacturer

- Before starting any body mounting work, the body manufacturer must check whether
 - The vehicle is suitable for the planned body,
 - The vehicle model and equipment remain suitable for the operating conditions even after the body has been installed.
- For the planning of add-on equipment, 3D design data, 2D drawings (offer drawings), product information and technical data can be requested from the responsible department or accessed via the communication system, see Chapter 2.1 and 2.2 of the corresponding mounting directive.
- Also note the special equipment offered ex works and prioritize and use it when choosing the basic vehicle.
- Vehicles delivered ex works comply with EC directives and country-specific regulations. This does not apply to vehicles without or with incomplete CoC documents or country-specific registration documents and, depending on the vehicle equipment, in some cases also to vehicles for non-European countries.
- The vehicles must still comply with EC regulations and national regulations after modifications have been carried out.
- The bodybuilder manufacturer must inform the officially recognized approval official or inspector of any modifications to the basic vehicle. Acceptance and inspection organizations will decide on compliance with legislation and regulations after any modifications have been made to the basic vehicle, and therefore on the eligibility for registration of the complete vehicle.
- The bodybuilder manufacturer must ensure that the vehicle meets the registration requirements with its modifications.

! NOTE

Observe all national laws, directives, and registration regulations.

1.3.2 Conventions

The layout and conventions used in this document are the same as those of the mounting directives.

3.3 Selecting the basic vehicle

Vibration absorber

In addition to chassis vehicles with the "Mandatory box body by bodybuilder manufacturer" equipment (code X4P) described in the Supplemental Directive for the eSprinter Model Series 907 YoM2025-1a, Bodybuilder Information Bulletin 03/2025 informed of the following change:

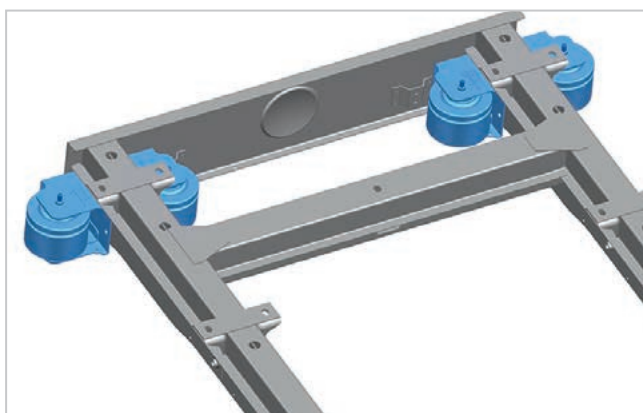
From production start in the 2nd quarter of 2025, chassis vehicles will also be available without equipment code X4P in a vehicle configuration with or without a platform ex factory (code P02).

- i** You can find further information about the availability of this vehicle configuration from your Mercedes-Benz AG Sales Partner.

! NOTE

As an additional measure, a vibration absorber is installed in these vehicles on both sides on the left and right of the longitudinal frame members in the rear area. See the following figure. A change to these component parts, their position or fastening, or their removal is not permitted.

Both the effects of additional component part weights (total approx. 40 kg) on the remaining available payload and the effects of the installation space occupied on the assembly of detachable parts in the rear area must be taken into consideration when planning respective conversions or bodies.



Vibration absorber on rear longitudinal frame member (schematic)

- i** You can find details about the vibration absorbers in the 3D design data. You can find information on the provision of 3D design data in the Bodybuilder Portal in Chapter 2.2.1 of the corresponding mounting directive.

Vibration absorbers are not installed in combination with the following equipment:

- Code X4P "Mandatory box body by bodybuilder manufacturer"
- Code X2R "Mandatory camper van conversion by bodybuilder manufacturer"
- Code PV1 "Pre-installation for retrofit of cargo lift-gate"

! NOTE

When planning bodies and modifications, you should consider that chassis vehicles with pure platform bodies are not permitted without vibration absorbers. This excludes vehicles with platform and tarp bodies in combination with a cargo liftgate.

- i** If you have questions or desire further information about your specific body or conversion, then you should contact the Technical Consultancy Team, see Chapter 2.1 of the corresponding mounting directive.

8.4 Interfaces and electric lines

NOTE

If work is carried out on a vehicle with an electric drive, country-specific specifications regarding statutory protective equipment and accident prevention regulations must be observed, e.g. the latest audit standard of DGUV Information 209-093 "Training for work on vehicles with high-voltage on-board electrical systems" for Germany.

Always have work on electric and electronic devices performed by an electrician or at a specialist workshop qualified for the eSprinter model series 907. It is recommended that you use a Mercedes-Benz Service Partner for this.

NOTE

Carefully read through the information in the Supplemental Directive for the eSprinter Model Series 907 as a supplement to the current change information and observe all of the specifications for work on a vehicle with an electric drive.

8.4.8 Current collection

In addition to the two familiar methods for current collection for electrical consumers specific to the bodybuilder manufacturer, the electric power take-off will in future provide a further special equipment option in the eSprinter model series 907 with electric drive:

Code	Equipment ¹⁾	Current collection point	Information and specifications
EK1	Terminal strip	In driver's seat box	See Mounting Directive for Sprinter Model Series 907 and Supplemental Directive for eSprinter Model Series 907
E2I	Auxiliary battery in the interior	Fuse housing (MFB ³⁾) in driver's seat box	
E2M	Auxiliary battery in the engine compartment		
E2C	ePTO ²⁾ electric power take-off	Fuse housing (MFB ³⁾) in engine compartment	Current change Information 01 / 2025

- 1) The description is intended as an explanation and is not the same as the code designation
- 2) ePTO: Electric power take-off
- 3) MFB: Main fuse box

- i** The auxiliary batteries (code E2M and E2I) and cut-off relay (code E36) are not available for the retrofit of auxiliary batteries specific to the bodybuilder manufacturer with code E2C.
- i** Note the special Bodybuilder Manufacturer Information Bulletin on the availability of special equipment code E2C. You can find further information from your Mercedes-Benz Sales Partner.

Electric power take-off (code E2C)

The electric power take-off (ePTO) is power supply from the high-voltage battery through a voltage converter (DC/DC converter). Current (LV+ / 12 V DC) is collected at four connection points in a main fuse box (MFB) fuse box in the engine compartment for the connection of electrical consumers specific to the bodybuilder manufacturer.

The output data for this electrical interface is listed in table (→ page 9).

Either the ground bolts in the vehicle can be used to connect the electrical ground (LV-) or the bodybuilder manufacturer can independently attach additional ground bolts to suitable areas of the body-in-white. Observe further information and specifications in Chapter 8.4.12 of the respective mounting directive.

The ePTO is activated and deactivated using a switch in the cockpit of the vehicle. The option for current collection is available during driving operation, during parking and during the charging process or preconditioning of the high-voltage battery. It is also possible to leave and lock the vehicle while the ePTO is still in operation. The operating states of the ePTO are indicated by an LED in the switch and through different display messages in the instrument cluster (see Operator's Manual) and are described below.

-  You can find further information and specifications for use of the ePTO in the Operator's Manual for your vehicle.
-  You can contact the Technical Consultancy Team if you have any questions, see Chapter 2.1 of the corresponding mounting directive.

 NOTE

Current can only be collected through the connections of the MFB in the engine compartment that are intended for the bodybuilder manufacturer and represented in the following figure.

Any change to the components of the ePTO installed ex factory, including the lines that are already installed, is not permitted.

Electrical consumers may not be connected to the buffer battery necessary for operation of the ePTO in the engine compartment, nor may any changes be made to it.

It is not permitted to use the buffer battery for a jump start.

 WARNING

A connection between the current collection points of the MFB of the ePTO (code E2C) and the 12 V on-board electrical system of the vehicle (e.g. terminal strip code EK1) is not permissible under any circumstances.

If this is not observed, the components of the ePTO or the on-board electrical system and individual control units of the vehicle systems may be damaged or malfunction.

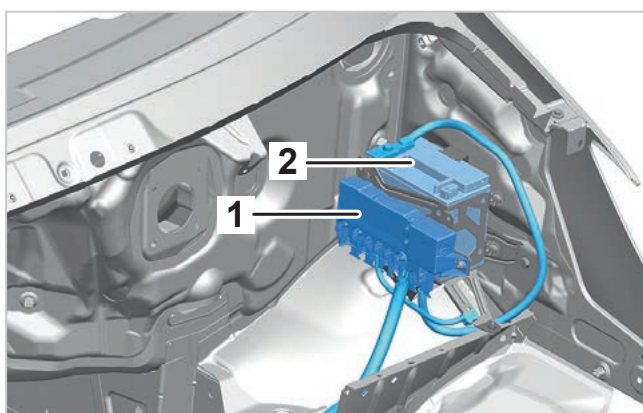
There is risk of an accident and danger to life and limb!

! NOTE

Connecting additional voltage converters (DC/DC or DC/AC converters) to the ePTO can lead to negative impacts on the current collection or on vehicle systems in isolated cases due to various electrical influence factors.

If you plan to use voltage converters, please contact the Technical Consultancy Team during the planning phase. A Certificate of Compatibility may be necessary (see Chapter 2.1 of the corresponding mounting directive) and this can be requested through the Bodybuilder Portal (see Chapter 2.2.1 of the corresponding mounting directive).

- Remove MFB cover before connecting electrical consumers and reattach it again after installation work so that electrical connections are protected from external influences. Replace a damaged MFB cover if needed.
- The bodybuilder manufacturer is responsible for ensuring that electrical installation work, including line installation and fastening in the vehicle, is conducted by a qualified electrician. The information and specifications on this in the corresponding mounting directives and supplemental directives must be observed.



Components of the ePTO in the engine compartment (schematic)

- 1 MFB fuse box without cover
2 Buffer battery (12 V)

! NOTE

The following output data and specifications must absolutely be observed. It is not permissible to exceed the limit values. Otherwise, the ePTO may be overloaded and fail, affecting the function of the overall vehicle.

The bodybuilder manufacturer is responsible for ensuring that these limit values are maintained through suitable line layout, function control and through safeguarding the connected electrical consumers.

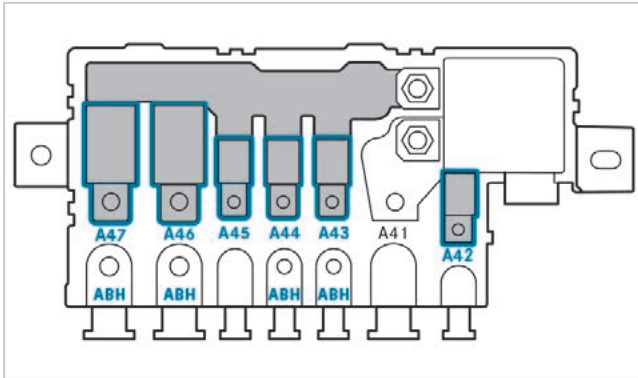
Additionally, the specifications on electromagnetic compatibility (EMC) in Chapter 8.2 of the corresponding mounting directive must be observed.

ePTO output data

Maximum continuous output¹⁾²⁾	max. 3.5 kW	-
Nominal voltage²⁾	14.3 V	Possible deviation depending on general conditions (state of charge, driving situation, ambient conditions, etc.)
Typical voltage range²⁾	13 - 15 V	Operational activity of connected consumers
Maximum current-carrying capacity¹⁾²⁾	max. 265 A	Sum of individual currents

- 1) Total for all connected electrical consumers.
2) Without taking into consideration line losses to the electrical consumer.

Current collection for bodybuilder manufacturer



MFB fuse box in engine compartment for ePTO

- MFB connections A43, A44, A46, A47 are permitted for use by bodybuilder manufacturers
- MFB connections A42, A41, A45 may not be used and may not be changed.

MFB connection	Fuse [A]	Cable cross-section (recommendation) [mm ²]	MFB terminal stud/tightening torque
A43	100	25	M6 × 20.3 mm / 8 ±0.8 Nm
A44	60	16	
A46	150	50	M8 × 25.3 mm / 16 ±1.6 Nm
A47	400	50	

i See Chapter 8.4.2 in the corresponding mounting directive for further information and specifications on recommended cable cross-sections.

! NOTE

Electrical loads caused by the connection of electrical consumers must fulfill the specifications from ISO 16750-2 that form the basis for the design of the ePTO. Electrical loads that deviate from this and could trigger behavior outside the limits of this standard are not permitted. If this is not observed, the ePTO may temporarily shut off or the ePTO may become damaged.

Non-linear loads must ensure that the maximum system ripple is not exceeded:

Frequency range [kHz]	Amplitude of the maximum permissible ripple ¹⁾
0.015–1	1.1 V _{p-p}
1–30	Linear collection: 1.1–0.212 V _{p-p}

1) Ripple V_{p-p}: "Peak-to-peak" voltage

Additionally, no load increase speed may be greater than the following limits during operational transitions (on/off and internal load changes):

Type	Slope rate [V/ms]	Maximum voltage pulse [V]
Rise	1	<1.1
Fall	1	<2.0

The actuation of loads with steep ramps, a high switch-on current (short circuit) and hard shut-offs can lead to the ePTO switching off.

Important information on usage of the ePTO

- Performance monitoring is integrated for compliance with the maximum continuous output (total of all consumers). If the limit values are exceeded, the ePTO will be deactivated and a display message will be shown in the instrument cluster.
Once the output need has fallen below the limit value, the ePTO can be activated again.
- During continuous operation, automatic shut-off occurs after 12 hours. This time monitoring is reset once the vehicle starts (Ignition Switched Off/On) and renewed activation of the ePTO is possible.
- If use of the ePTO has to be restricted by the system due to a low state of charge in the high-voltage battery with low range, the driver will be informed of this by a display message in the instrument cluster in two stages:
 - 1. Stage: Manual deactivation possible using a switch in the cockpit
 - 2. Stage: Automatic deactivation will occur after 2 minutes
- Renewed activation of the ePTO is possible while the high-voltage battery is charging as soon as the 2nd Stage display message is no longer shown.
- At ambient temperatures of more than 30°C, it is recommended that the continuous output necessary for the operation of electrical consumers be restricted to a maximum of 2.5 kW so that components do not overheat. A higher output is only permissible for electrical consumers with a brief operating duration.
- In ambient conditions below -15°C, which can lead to a temperature too low for the high-voltage battery, activation of the ePTO can only take place once the system has preconditioned (heated) the high-voltage battery.

 You can find details about the operating statuses and important information about use of the ePTO in the Operator's Manual for your vehicle.

 You can contact the Technical Consultancy Team if you have any questions, see Chapter 2.1 of the corresponding mounting directive.

8.9 Driving assistance systems

- i** You will be informed about changes and supplements to the pre-installation of the digital reversing camera (code FR7) in this change information. The entire chapter is listed for a complete description of the reversing camera.

8.9.9 Reversing camera

Available special equipment

Code	Equipment ¹⁾
FR3	Analog reversing camera with inside rearview mirror display
FR8	Digital reversing camera with display via central display
FV1	Pre-installation for analog reversing camera with inside rearview mirror display
FR7	Pre-installation for digital reversing camera with display via central display

- 1) This description is intended as an explanation and is not the same as the code designation.

! NOTE

The information and specifications in the Operator's Manual for your vehicle must be adhered to. Make yourself familiar with the functions of the driving assistance systems.

! NOTE

External contours of the vehicle body, detachable parts that protrude from the outer surface of the body, or retrofitted vehicle equipment can all lead to a restriction of the viewing area behind the vehicle or disturbances in the visual impression in the display, and thus result in limitations to the function of the driving assistance system.

! NOTE

Compliance with country-specific laws and guidelines on the viewing area behind the vehicle (e.g. UN R 158) and the fulfillment of the obligation to provide evidence for the registration and driving operation of the converted vehicle are the sole responsibility of the bodybuilder manufacturer.

! NOTE

Bodybuilder manufacturer-specific operator's manuals must contain appropriate warnings and information for the vehicle user, which at least refer to:

- Possible restriction due to retrofitted detachable parts or vehicle equipment that protrude into the viewing area of the reversing camera, e.g. bicycle rack at the rear of the vehicle;
- Deactivation of pedestrian emergency braking when reversing (→ page 15) and Trailer View (→ page 15), as well as hiding of reversing camera guide lines (→ page 14) after documentation of code O1N in the VeDoc vehicle data card;
- Possible restriction of the function of the driving assistance system if the reversing camera and its surroundings on the vehicle are damaged;
- Obligation to have the function of the driving assistance system checked if damage is detected and after accidents affecting the reversing camera or its surroundings on the vehicle; it is recommended to have this check carried out by a Mercedes-Benz Service Partner.

1) Reversing camera ex factory

Two reversing camera systems are available ex factory, which, depending on the country-specific legal regulations, are either already installed in the vehicle as standard or are available as special equipment:

- Analog reversing camera with inside rearview mirror display (code FR3)
- Digital reversing camera with display via the central display (code FR8)

If the parking package special equipment (code JB6 or JB7, see Chapter 8.9.8 of the corresponding mounting directive) is not selected for closed model designations, the digital reversing camera (code FR8) is fitted as standard in markets with CoC documents. The camera is also installed in these markets on open model designations with a platform ex works (code P02).

For other markets, the analog or the digital camera is available as special equipment depending on the country.

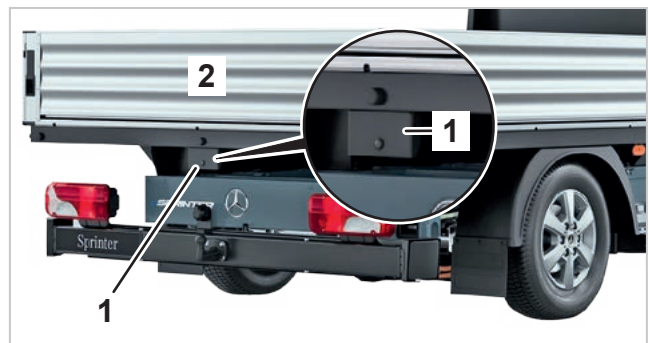
On closed model designations (panel van and Tourer), the installation is implemented with a holder on the rear roof frame; on open model designations with a platform ex factory (code P02) it is implemented with a holder under the platform.

Due to the installation space situation, the Trailer View function is deactivated in open model designations with platform ex factory (code P02) and is therefore not available in the head unit display.



Position of reversing camera on closed model designations

- 1 Holder for reversing camera on rear roof frame



Position of reversing camera for open model designations (example for vehicle with platform ex works)

- 1 Holder for reversing camera under platform
2 Platform ex works (code P02)

For open model designations without platform ex works (without code P02), pre-installations are available for the analog or digital reversing camera (code FV1 or FR7), see section 2 (→ page 15).

WARNING

Modifications to the camera systems and their surroundings as well as installation work affecting the function and detection range of the camera are not permissible.

Otherwise, driving assistance systems may no longer function as intended and legal requirements may no longer be met.

There is risk of an accident and danger to life and limb!

The bodybuilder manufacturer must clarify possible exceptions to detachable parts at the rear of the vehicle (e.g. bicycle rack) with the Technical Consultancy and/or the responsible registration authority with regard to the country-specific legal requirements (e.g. UN R 158).

In addition, it should be noted that detachable parts on the rear of the vehicle can lead to restrictions in the visibility area behind the vehicle and pedestrian emergency braking when reversing.

In the case of bodybuilder manufacturer-specific special attachments with a greater effect on the detection range of the camera, e.g. cargo liftgate, it is permissible to change the camera position in the longitudinal direction of the vehicle. The camera position and orientation can also be adapted within the limit values prescribed for the pre-installation (code FR7), see specifications in section 3 (→ page 16) and figures (→ page 19).

NOTE

The guide lines of the digital reversing camera (code FR8) in the head unit display are designed exclusively for the basic vehicle ex factory.

If approved detachable parts are retrofitted to the rear of the vehicle, the appropriate parameter set must be coded in the reversing camera by your Mercedes-Benz Service Partner and the guide lines must be hidden.

The guide lines of the reversing camera must also be hidden in the following cases:

- if the vehicle is extended to the rear by bodybuilder manufacturer-specific detachable parts,
- if the camera position is changed for special bodies,
- if the camera is retrofitted with the pre-installation code FR7, see section 2b (→ page 16).

To do this, the code O1N "Activation of reversing camera without guide lines" must be documented in the VeDoc vehicle data card.

In addition, if the position of the camera changes, the new position information must be added to the vehicle data card via XENTRY with the corresponding O-codes; see section 5 (→ page 28). The reversing camera must then be recoded and calibrated through XENTRY, see "Commissioning of Control Units" guide in the Bodybuilder Portal and in Chapter 8.19/8.18 of the corresponding mounting directive.

⚠ WARNING

For the previously mentioned body situations, it is not possible to ensure that the pedestrian emergency braking function works properly when reversing.

There is risk of an accident and danger to life and limb!

For this reason, the documentation of code O1N deactivates pedestrian emergency braking when reversing. Additionally, the Trailer View function is deactivated with code O1N and is therefore no longer available in the head unit display.

The operator's manual specific to the bodybuilder manufacturer must inform the vehicle user of the deactivation of this function and the fact that the guide lines are hidden (→ page 12).

2) Retrofitting the reversing camera

There are two options available for retrofitting a reversing camera on open model designations:

- Pre-installation for analog reversing camera with inside mirror display (code FV1)
- Pre-installation for digital reversing camera with display via the head unit display (code FR7)

! NOTE

Compliance with country-specific laws and guidelines on the viewing area behind the vehicle (e.g. UN R 158) and the fulfillment of the obligation to provide evidence for the registration and driving operation of the converted vehicle are the sole responsibility of the bodybuilder manufacturer.

2a) Analog reversing camera: Pre-installation of reversing camera with inside rearview mirror display (code FV1)

It is recommended to use the analog camera approved by Mercedes-Benz AG. It is not supplied with the vehicle on delivery and is therefore available as an individual part with an adapter wiring harness from the Mercedes-Benz Service Partner:

Item number **A 000 905 68 15**

- ① Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.
- ① Information on component geometry and the camera's cone of view can be found under the item number in the 3D design data and 2D drawings. You can find information on the provision of 3D design data in the Bodybuilder Portal in Chapter 2.2.1 of the corresponding mounting directive.
- ① For special bodies and questions, please contact us by e-mail in German or English, stating the vehicle model designation:

bb-3d_data-van@mercedes-benz.com



! NOTE

If you are using analog cameras other than those approved by Mercedes-Benz AG, these must comply with the same standard NTSC 60 Hz.

With this pre-installation, an interface for connecting the analog reversing camera is available for open model designations in the driver's seat box and for closed model designations in the area of the third brake lamp (roof in the upper rear area), see section 4 (→ page 22).

! NOTE

The reversing camera system is classified as ASIL Level A according to ISO 26262. For all the components involved, it must be ensured that no delayed or frozen images can occur, see section 5 (→ page 28). The bodybuilder manufacturer must provide proof of this on request.

2b) Digital reversing camera: Pre-installation for electric reversing aid (code FR7) for retrofitting of a digital reversing camera with display via head unit display

This special equipment is only available for open model designations. For closed model designations, the special equipment code FR8 must be used; see section 1 (→ page 13).

! NOTE

Only the digital reversing camera approved by Mercedes-Benz AG must be used for the digital reversing camera system.

Any other digital camera is not permitted, because it is not approved for combining with the other system components (e.g. the central display).

Code FR7 is only possible in combination with the MBUX multimedia system with 10.25-inch touchscreen (code E7M or E7H).

This pre-installation provides the interface for connecting the reversing camera in the driver's seat frame; see section 4 (→ page 22).

3) Retrofitting the digital reversing camera with pre-installation code FR7

The digital reversing camera is not included when the vehicle is delivered and is therefore available (with hardware and software in the delivery package) without wiring harness via a Mercedes-Benz Service Partner:

Item number **A 000 900 38 55**

- i The digital reversing camera for code FR7 was updated to the above item number with a software change (Release 12).
 Item number A 000 900 85 50 can still be used because the previous software (Release 10) will automatically be updated to the new release (Release 12) during the initial commissioning. A subsequent software update is also possible through XENTRY.

NOTE

For cameras with the new software (Release 12), observe the changed specifications for installation of the camera with regard to height above the roadway and angle in the "Specifications for Positioning and Installation" section (→ page 18) and commissioning in Section 5 (→ page 28).

The specifications in the Mounting Directive Edition YoM2025-1a are still valid for cameras with the earlier software (Release 10).

The 3D design data of the camera with the 3D cone of vision is available for the digital planning and design of a holder for attachment of the reversing camera to the body:

Item number **A 177 901 06 04**

- ① Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.
- ① Information on component geometry and the camera's cone of view can be found under the item number in the 3D design data and 2D drawings. You can find information on the provision of 3D design data in the Bodybuilder Portal in Chapter 2.2.1 of the corresponding mounting directive.
- ① For special bodies and questions, please contact us by e-mail in German or English, stating the vehicle model designation:

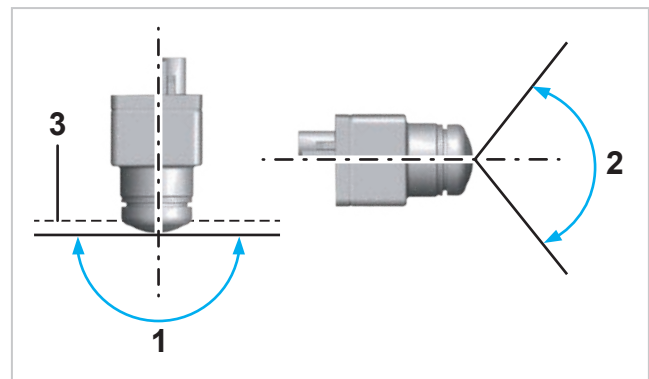
bb-3d_data-van@mercedes-benz.com

**Specifications for the reversing camera holder**

The bodybuilder manufacturer must produce a separate holder for positioning and installing the reversing camera, which must meet the following specifications:

- The holder must be specially adapted for installation on the bodybuilder manufacturer's specific body and be suitable for use of the camera while driving.
- The holder must not restrict the viewing area or the viewing angle limits of the camera. For this, it is essential to make the edge of the holder in the directly adjacent area of the camera lens without any protrusion; see following illustration.

For the design of the holder, use the previously mentioned 3D design data of the cone of vision.



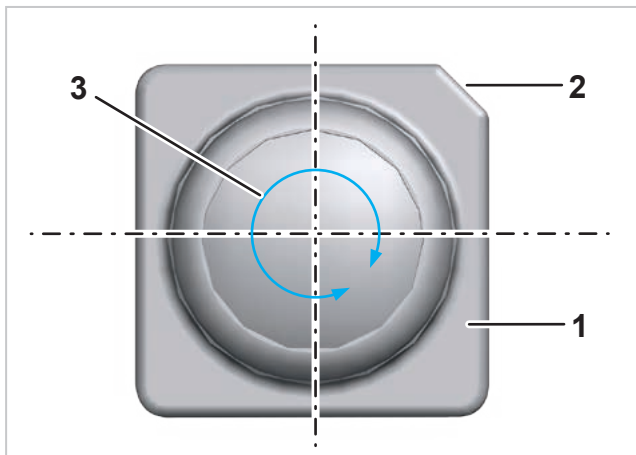
Horizontal **1** and vertical **2** viewing area or viewing angle limit and limit contour for camera holder **3** (schematic); see 3D design data for details of the viewing cone

- ① The holders of the reversing camera for closed and open model designations ex factory with code FR3 or FR8 (→ page 13) can potentially be used for installation on the bodybuilder-manufacturer-specific body or as a basis for the structural design of your own holder (→ page 21).

! NOTE

The camera must be attached with the holder so that the beveled corner is exactly at the top right (when viewed from behind), see the following illustration.

A maximum deviation in the installation position of $\pm 1^\circ$ (rotation around the center axis of the camera) is permitted.



Alignment of the reversing camera on the body, view from behind

- 1 Reversing camera (RCV5)
- 2 Beveled corner at top right
- 3 Permissible tolerance of the installation position around the axis of rotation: max. $\pm 1^\circ$

- For consistently good image quality, design the holder with high component rigidity to reduce camera vibrations as much as possible.
- Comply with IP protection class IP6K7K for the camera and electrical connections.
- Observe the electrical connection specifications (→ page 22) also when planning, manufacturing and installing the holder.
- The plugs of the cables must be located in the interior of the body.

- Take precautions to ensure that no liquid collects in the holder or in the cut-out in the body
- Take suitable measures to protect the line from damage in the area of the cable run into the body interior. Also seal the cable run against water and dirt ingress.

When using the adapter wiring harness, a variant with cable grommet is available, see section 4b (→ page 26). This seals the opening of a cable run with a sheet metal part.

Positioning and installation specifications

! NOTE

The limit values for the installation position and alignment of the reversing camera on the vehicle rear must be observed.

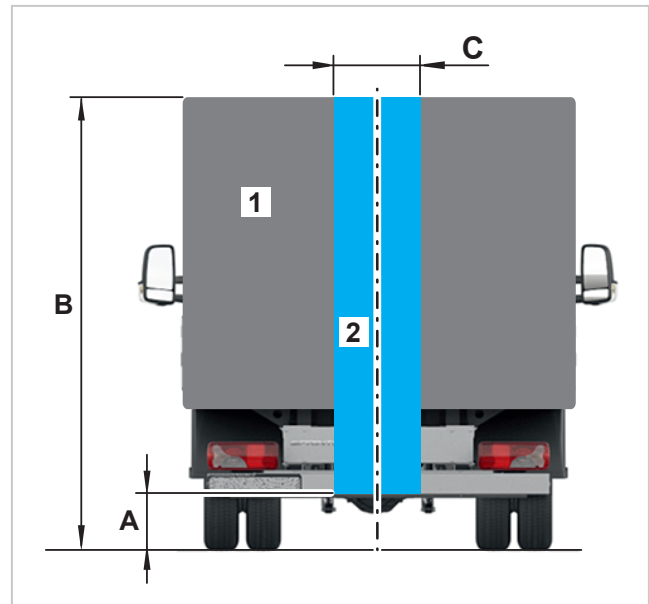
A Certificate of Compatibility may be necessary and this can be requested through the Bodybuilder Portal, see Chapter 2.1 and 2.2.1 of the corresponding mounting directives. Also observe the information in section 2 (→ page 15).

- The reference point for positioning the camera is the center of the camera lens.
- Depending on the bodybuilder manufacturer-specific body, a position in the marked area **2** of the rear of the vehicle can be selected for the installation of the camera.
- Positioning and installation of the reversing camera on the upper roof frame is permitted if the limit value for the maximum height is not exceeded, see the following illustration (→ page 19).

- If the reversing camera is installed directly on the body in the case of bodybuilder manufacturer-specific special bodies and if there is a greater restriction of its recording range, which is no longer permissible according to the country-specific regulations, it is permissible to change the camera position in the longitudinal direction of the vehicle (X-axis on the vehicle side). An adapter component part for bridging the distance between the camera and the body must be sufficiently rigid and stable for driving operation. In the case of special bodies like this, it is recommended that you contact the Technical Consultancy Team for Bodybuilder Manufacturers during the planning phase in order to coordinate the planned body work and its compatibility with the driving assistance system.

The guide lines of the reversing camera must then be hidden during commissioning with code O1N; see note at the end of section 1 (→ page 13).

- The installation position of the camera in the Z direction and X direction must be specified in the input screen in the Bodybuilder Portal prior to commissioning to ensure correct coding of the control unit with O-codes and calibration of the camera; see the following figure with details on the permissible range for the installation position, as well as the information in Section 5 (→ page 28).
- An incline angle dependent on the installation height is specified for the camera with new software (Release 12) for optimum display in the head unit display, see figure and table (→ page 20). The incline angle must be entered in XENTRY during commissioning for calibration of the camera.



Permissible range for the installation position of the reversing camera in the Y and Z directions (schematic)

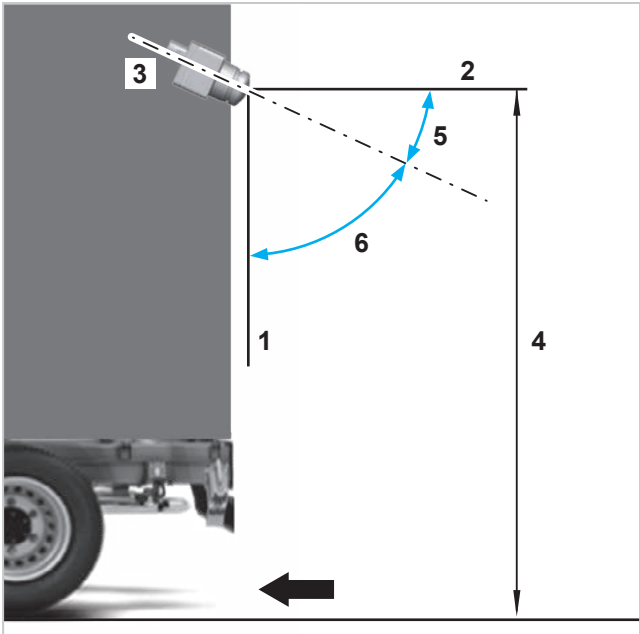
- 1 Example of any bodybuilder manufacturer-specific body
- 2 Permissible range for the installation position

Permissible range for the installation position

	Installation position	Limit value [mm]
A	Smallest height above roadway ¹⁾	540 ²⁾
B	Greatest height above roadway ^{1) 3)}	4000 ³⁾
C	Maximum deviation, in relation to vehicle center	±200

- 1) The height in the Z-direction above the roadway must be measured on the vehicle with a fully completed body in running order without load.
- 2) Depending on the type of body and the installation position of the camera, the legal requirements (e.g. as per UN R 158) may not be fulfilled at low height levels, particularly between 540 and 700 mm. Check the prerequisites for compliance with legal requirements for the body-specific installation situation.
- 3) Limit value is only valid for new software (Release 12). Maximum of 3500 mm permissible with old software (Release 10).

The installation position in the X direction is entered prior to commissioning as the distance from the center of the front axle (differentiation up to 5900 mm or from 5901 mm), see Section 5 (→ page 28).



Alignment of the camera on the body (schematic)

- 1 Vertical Z-direction
- 2 Horizontal X-direction
- 3 Center axis of camera lens
- 4 Position in Z-direction: Installation height above roadway
- 5 Angle of inclination to the horizontal plane (for entry in XENTRY)
- 6 Angle of inclination to the vertical plane (for ease of measurement on the body)
- ➔ Direction of travel

Inclination alignment of camera

4 Height above roadway ¹⁾ [mm]	Basic setting of incline angle [°]	
	5 Angle to the horizontal ²⁾	6 Angle to the vertical ³⁾
540–700	7	83
701–1000	15	75
1001–1500	25	65
1501–2000	35	55
2001–2500	45	45
2501–3000	50	40
3001–3500	55	35
3501–4000 ⁴⁾		

- 1) Observe information on installation position in Z direction (→ page 19)
- 2) Angle for entry into input screen in Bodybuilder Portal or in XENTRY
- 3) Angle for simpler measurement relative to body
- 4) Height level is only valid for new software (Release 12). Maximum of 3500 mm permissible with old software (Release 10).

- A deviation from the basic setting of the incline angle by **maximum $\pm 2^\circ$** is permissible for further optimization of the camera image in the head unit display. This specification also applies when reusing existing camera holders.

- When commissioning and calibrating the reversing camera using XENTRY, the digital field of view in the head unit display can be adjusted by an additional $\pm 3^\circ$.

Exceeding the total tolerance of $\pm 5^\circ$, however, leads to entries in the fault memory in XENTRY and is therefore not permissible.

Observe national legal specifications when setting the field of view, e.g. as per UN R 158, see note at the beginning of the chapter (→ page 12).

- If the specifications on the incline angle **cannot** be observed in the following exceptional cases, software adaptation to the old version (Release 10) can be achieved by documenting O-code O1X "Generation 1 camera holder" during commissioning.
For this, the new image optimization from the software conversion (Release 12) will not be implemented and the installation position in the Z direction may be a maximum of 3500 mm.

The following exceptional cases are permitted:

- Reusing the previous camera holder, e.g. if just the camera is being replaced as the result of damage
- Design and use of a new camera holder with an adjusted incline angle is not practicable, e.g. due to restricted installation space or integration into the overall body
- The angle of inclination must be specified for the calibration of the camera during commissioning in XENTRY; see section 5 (→ page 28). After changing the angle of inclination, the changed value must be entered again and the camera must be recalibrated.
- Vibrations and movements of the camera while driving can have a negative effect on the image quality and must therefore be reduced as much as possible. It is recommended that the holder with camera only be installed on structural components and/or in vehicle surroundings with high inherent rigidity.
- When positioning and installing the camera, ensure that the viewing area to the rear is not restricted. Observe the notes on impermissible restrictions of the viewing area, see Chapter 8.9.9 (→ page 12) and the legal requirements for the viewing area and permissible interfering contours according to Section 2 (→ page 15).
With regard to this, also note the illustration (→ page 17).

Possible use of standard holders (from code FR8)

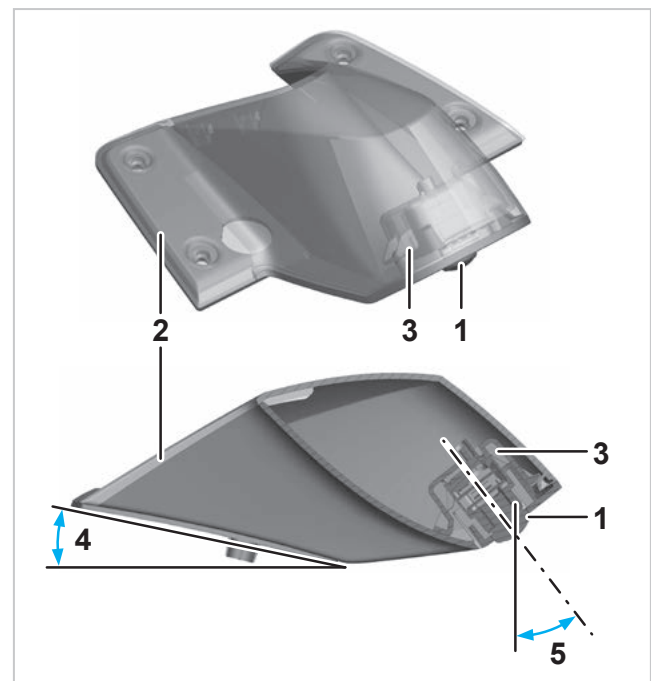
a) Closed model designations

The reversing camera (code FR8) is installed ex works on the closed model designation with a holder on the rear roof frame with roof height code LH2, see figure (→ page 13).

This holder can also be used for a bodybuilder manufacturer-specific body solution or as a basis for the design of a new holder.

Item number **A 907 820 93 01**

Note that this holder has an inclination **4** and a curvature on the underside due to the vehicle contour and that the camera is attached to the holder with a tilt to the vertical plane **5**. This must be taken into account when planning to use the holder.



Reversing camera holder (code FR8)

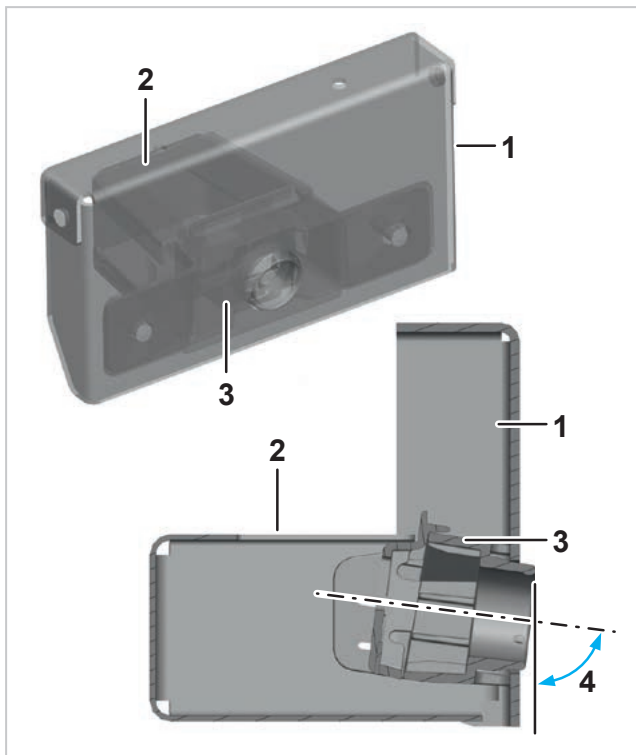
- 1 Reversing camera
- 2 Holder
- 3 Camera mounting
- 4 Inclination of holder approx. 13° in CAD design position
- 5 Camera tilt approx. 40° in CAD design position

b) Open model designations

Ex factory (code P02), the reversing camera (code FR8) is installed on a structural part (retaining plate) underneath the platform with fastening parts and covers on the open model designations; see illustration (→ page 13).

These component parts can be used for a bodybuilder manufacturer-specific body solution or as a basis for the design of a new holder.

Make sure that the camera is mounted in the holder with a tilt to the vertical **4**. This must be taken into account when planning to use the holder.



Mounting and cover parts for reversing camera

- 1 Holder plate under platform
- 2 Camera cover with electrical connection
- 3 Camera attachment (illustration without reversing camera)
- 4 Camera tilt approx. 82° in CAD design position

i Details of the holders of the reversing camera are available in the 3D design data and 2D drawings. You can find information on the provision of 3D design data in the Bodybuilder Portal in Chapter 2.2.1 of the corresponding mounting directive.

i Item numbers for other versions of these holders, e.g. variants depending on the steering type or in combination with the rear-view camera for digital inside mirror (code F84/F85, see Chapter 8.9.10 of the corresponding mounting directive) as well as for fastening parts and covers for the camera for open model designations with platform ex factory (code P02) can be obtained from the Technical Consultancy Team, see Chapter 2.1 of the corresponding mounting directive.

i Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.

4) Electrical connection

After installation, the reversing camera must be electrically connected directly to the vehicle's connecting point by a qualified electrician using a video line and, in the case of the analog camera, also using a power supply line.

! NOTE

For the production of a video line (high-frequency coaxial line), a wiring harness manufacturer is recommended that is qualified for FAKRA connections (plugs or sockets) and has the necessary production facilities.

The wiring harness must satisfy the following requirements:

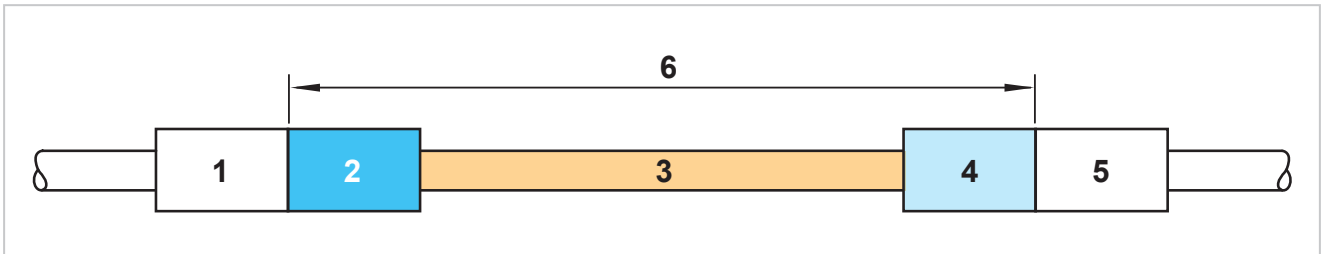
- The wiring harness (cable with plug) made by a qualified electrician or wiring harness manufacturer must be checked for electrical connections and functionality before use (according to DS characteristic: Mandatory documentation relevant to safety). The test results must be submitted on request.
- An additional connecting point along the length of the line between the camera or adapter wiring harness (if used) and the vehicle's connecting point is not permitted.
- The bodybuilder manufacturer must ensure a watertight connection of the lines to the camera.
- When laying the video cable, the minimum bending radius must not be less than $5 \times d$ (d = cable diameter).
- Ensure that lines are routed correctly in the vehicle, see Chapter 8.4 of the corresponding mounting directive.
- The quality of the video/high-frequency line must correspond at least to the values specified by LEONI Dacar.
- The line quality was designed for the lengths of line specified in sections 4a (→ page 24) and 4b (→ page 26) from the connecting point to the camera. If a longer line is required, its functional capability must be assured by means of tests conducted at the bodybuilder manufacturer's expense. The test results must be submitted on request.
- The reversing camera system must not be disrupted by EMC influences, e.g. by 230 V systems, actuators etc.

4a) Analog reversing camera (code FV1)

The position of the connecting point on the vehicle for the electrical connection is located:

- on open model designations in the driver's seat box,
- on closed model designations in the area of the third brake light (roof area at the rear of the vehicle).

Observe the following information on the necessary parts for the production of a wiring harness:



Assignment of the line components to the following tables

For information only (**white**):

- 1 Connecting points on the vehicle
- 5 Connections on the adapter wiring harness of the camera

Required for the production of the wiring harness 6:

- 2 Connections to the connecting point (**blue**)
- 3 Electric line (**orange**)
- 4 Connections to the adapter wiring harness (**light blue**)

Electric line

	Type	Cross-section or specification	Manufacturer	Maximum permissible length
3	FLRY Voltage	2 × 0.5 mm ²	Commercial provider or cable manufacturer	10 m
	Video (coax)	RG179/Dacar 360-Coax-B (105)-75-1.68-2.7	Leoni	10 m

Observe all technical specifications and requirements in the manufacturer's data sheet.

Electrical contacts

	Position	Type	Type (pins)	Reference item number	Manufacturer	Manufacturer no. coding, model
1	Connecting points on the vehicle	Voltage	Socket (4 pins)	A 037 545 65 28	TE connectivity	1-929170-1 Coding A, female
		Video (Fakra)	Socket	A 023 545 60 26	Rosenberger	59Z063-DC0-F Coding F, female
2	Connections to the connecting point	Voltage	Plug (4 pins ¹⁾)	A 056 545 40 28	TE connectivity	1-2112850-2 Coding A, male
			Pin contact	A 032 545 39 28		963730-1
		Video (Fakra)	Plug	A 053 545 40 28	Rosenberger	59Z064-DC0-F ²⁾ Coding F, male
			Pin contact	A 001 982 07 28		59S17E-1D8A4
4	Connection to the power line of the camera	Voltage	Socket (2 pins ¹⁾)	A 029 545 10 26	Hirschmann	805-120-511 Coding A, female
		Contact	Socket contact	A 003 982 74 26	Kostal Kontakt Systeme ³⁾	22140734133
		Seal		A 001 545 50 80		10800507250
	Connection to the video line of the camera	Video (Fakra)	Socket	A 000 545 91 12	Rosenberger	59Z163-DC0-F ²⁾ Coding F, female
		Fakra crimp	Socket contact	A 001 982 10 26		59K17F-1D8A4
			Seal	A 000 542 81 00		59K14B-1M4/50
			End cover	A 000 553 00 00		59Z163-000/51
5	Connections on the adapter wiring harness of the camera	Voltage	Plug (2 pins)	A 211 545 15 28	Hirschmann	872-863-501 Coding A, male
		Video (Fakra)	Plug	A 000 545 93 12	Rosenberger	59Z176-C01-F Coding F, male

- 1) Pin assignment: Pin 1 GND, pin 2 UBAT (tml. 15)
- 2) Mercedes-Benz-specific parts with "DC0" (e.g. 59Z064-DC0-F) are available in neutral form from the manufacturer with "000" (e.g. 59Z064-000-F).
- 3) Sale via distributor, e.g. Herth+Buss

- ❗ Contact the parts manufacturer for alternative successor numbers if the part number listed is invalid or the parts cannot be delivered.
- ❗ You can contact the Technical Consultancy Team if you have any questions about electrical connection, see Chapter 2.1 of the corresponding mounting directive.
- ❗ Further information with illustrations of the connecting points can be found in the interface overview for BR 907 and BR 910 in the Bodybuilder Portal.

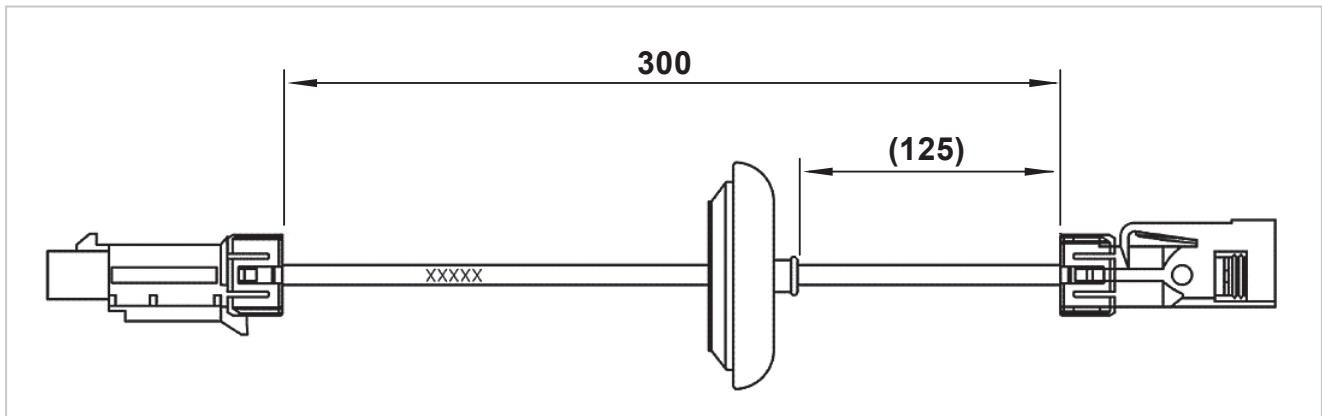
4b) Digital reversing camera (code FR7)

The following variants of a short, optional adapter wiring harness with a line length of 300 mm are available for easier assembly of the camera with holder on the body:

- With cable grommet:
Item number **A 907 820 83 01**
- Without cable grommet:
Item number **A 907 820 75 01**

 Contact your Mercedes-Benz Service Partner for information on successor numbers if the specified item numbers are invalid.

The cable grommet can be slid on the cable, is suitable for sheet metal parts from 0.6 to 3 mm and requires a round opening with a diameter of 25 mm (see 2D drawing of cable grommet A 000 998 55 02).



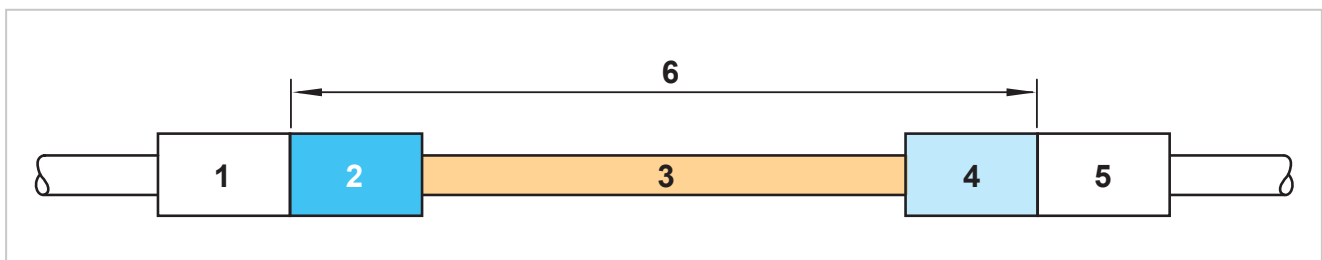
Adapter wiring harness for easier assembly (value in brackets depending on the sliding position of the cable grommet)

The position of the connecting point on the vehicle for the electrical connection is located:

- on open model designations in the driver's seat box.

For closed model designations, only the digital reversing camera is fitted ex works (code FR8), therefore no connecting point for pre-installation is available.

Observe the following information on the necessary parts for the production of a wiring harness:



Assignment of the line components to the following tables

For information only (**white**):

- 1** Vehicle-side connecting point
- 5** Connection on the adapter wiring harness or direct connection of the camera

Required for the production of the wiring harness **6**:

- 2** Connection to connecting point (**blue**)
- 3** Electric line (**orange**)
- 4** Connection to the adapter wiring harness or to the direct connection of the camera (**light blue**)

Electric line

	Type	Specification	Manufacturer	Maximum permissible length
3	Video (coax)	RG174/Dacar 462-Coax 50-1.52-2.8/T105	Leoni	14.5 m
	Alternative: Video (coax)	RTK031/Dacar 302-Coax-B(105)-50-2.1-3.3	Leoni	18 m

Observe all technical specifications and requirements in the manufacturer's data sheet.

Electrical contacts

	Position	Type	Type	Reference item number	Manufacturer	Manufacturer no. Coding, type
1	Vehicle-side connecting point	Video (Fakra)	Socket	A 023 545 60 26	Rosenberger	59Z063-DC0-F Coding F, female
2	Connection to connecting point	Video (Fakra)	Plug	A 053 545 40 28	Rosenberger	59Z064-DC0-F ²⁾ Coding F, male
	For line only RG174/Dacar 462	Fakra crimp parts set	Pin contact	A 000 982 02 28		59S100-102A4
	For line only RTK031/Dacar 302	Fakra crimp parts set	Pin contact	A 000 982 07 28		59S100-1M4A4
4	Connection to the adapter wiring harness or camera	Video (Fakra)	Socket (waterproof)	A 000 545 91 12	Rosenberger	59Z163-DC0-F ²⁾ Coding F, female
	For line only RG174/Dacar 462	Fakra crimp	Socket contact	A 000 982 47 26		59K130-102A4
			Seal	A 000 542 80 00		59K14B-102/50
			End cover	A 000 553 00 00		59Z163-000/51
	For line only RTK031/Dacar 302	Fakra crimp	Socket contact	A 000 982 73 26		59K130-1M4A4
			Seal	A 000 542 81 00		59K14B-1M4/50
			End cover	A 000 553 00 00		59Z163-000/51
5	Connection on the adapter wiring harness or direct connection of the camera ¹⁾	Video (Fakra)	Plug	A 000 545 93 12	Rosenberger	59Z176-C01-F Coding F, male

- 1) Integrated in camera
- 2) Mercedes-Benz-specific parts with "DC0" (e.g. 59Z064-DC0-F) are available in neutral form from the manufacturer with "000" (e.g. 59Z064-000-F).

❗ Contact the parts manufacturer for alternative successor numbers if the part number listed is invalid or the parts cannot be delivered.

❗ You can contact the Technical Consultancy Team if you have any questions about electrical connection, see Chapter 2.1 of the corresponding mounting directive.

❗ Further information with illustrations of the connecting points can be found in the interface overview for BR 907 and BR 910 in the Bodybuilder Portal.

5) Commissioning of the reversing camera

! NOTE

Installation of the digital reversing camera in combination with code FR7 must absolutely be documented in the VeDoc data card with O-code O1N "Reversing camera activation without guide lines" and the O-codes on the installation position, see following information and commissioning guide.

This is also absolutely necessary if the camera position is changed for code FR8, see note at the end of section 1 (→ page 13).

After installation of the reversing camera as per Section 3 (→ page 16) and its electrical connection to the vehicle's connecting point, commissioning must be performed via the Mercedes-Benz XENTRY Kit diagnosis tool before the vehicle is put into circulation.

- ① You can find the commissioning guide ("Commissioning of Control Units" guide) and access to the input screen for entering the values for the installation position in the Bodybuilder Portal:

<https://bb-portal.mercedes-benz-vans.com>



- ▶ After logging into the Bodybuilder Portal, select the "Order data" button.
- ▶ Select the order for your vehicle using the order number to access the specific order overview.
- ▶ Select the "Feedback data for commissioning" button.
- ▶ When commissioning for the first time, select the sub-item "Activate pre-installation". There, you will find the pre-installations ordered for the vehicle and the field for entering the camera position and angle of inclination.

! NOTE

After the vehicle has been put into operation for the first time or after a subsequent change (e.g. to the position of the camera), adjustment via the Bodybuilder Portal is no longer possible.

In this case, the O-codes for specifying the camera position and incline angle must be adjusted manually in the vehicle data card using XENTRY; the affected control units must then be re-coded and the camera recalibrated in accordance with the commissioning guide.

- ① You can find further information about this procedure in the commissioning guide and in Chapter 8.19/8.18 of the corresponding mounting directive.
- ① For information on XENTRY, see Chapter 2.2.4 of the corresponding mounting directive.
- ① It is recommended to have the commissioning performed by a Mercedes-Benz Service Partner.

Overview of the O-codes for commissioning the reversing camera (code FR7)

The O-codes assigned when entering the camera position are listed in the following tables for planning the body and determining the installation position. The values specified in these tables apply to vehicles with a fully completed body in running order without load.

 If you have any questions about the O-codes or if you cannot find the corresponding O-codes for your body situation in this overview, please contact the Technical Consultancy Team, see Chapter 2.1 of the corresponding mounting directive.

a) Position of camera in Z direction

O-code	Height above roadway [mm]
OZ1	540–700
OZ2	701–1000
OZ3	1001–1500
OZ4	1501–2000
OZ5	2001–2500
OZ6	2501–3000
OZ7	3001–3500
OZ8	3501–4000

b) Position of camera in X-direction

O-code	Distance from the center of the front axle [mm]
OX1	up to 5900
OX2	from 5901 onward

The O-code O1X "Generation 1 camera holder" must be documented in the data card of the vehicle using XENTRY for the exceptional cases described in Section 3 (→ page 21). This means that the previously listed O-codes will not be taken into consideration by the camera control unit.

After commissioning and before the vehicle is placed on the market, the reversing camera system must be tested under real conditions (including the bodybuilder manufacturer's requirements) and must function without errors. In addition to country-specific requirements, e.g. according to UN R 158, the following requirements must also be met:

- When reverse gear is engaged, the picture from the reversing camera is displayed completely and without errors in the central display.
- No guide lines are displayed.
- The rear of the vehicle is clearly visible.

NOTE

The reversing camera system is classified as ASIL Level A according to ISO 26262. For all the components involved, it must be ensured that no delayed or frozen images can occur. The bodybuilder manufacturer must provide proof of this on request.

